Pure Icons

Design guides

Designed by HASSEL

Introduction

What Is Pure Icons?

Pure Icons is an open-source icon project originally created for KLASSE OS. Its purpose is to establish an elegant, clean, and distinctive visual language for digital interfaces.

The design draws inspiration from both the past and the present of digital design. Two of our key sources of inspiration are the **Nokia MeeGo** icon system and **Apple's Liquid Glass** design language. Drawing from these different approaches, we seek to combine the simplicity of modern flat icons with the use of layering and visual depth.
Pure Icons does not aim to copy an existing style.

Our goal is to create an independent, consistent, and extensible icon system in which simple forms and refined visual elements come together as a cohesive whole.

Introduction

Purpose of This Guide

This Design Guide introduces the design philosophy behind Pure Icons, the rules governing icon creation, and the practical workflow used to produce icons.
Its purpose is to provide a consistent reference for using the icon set, designing new icons, and maintaining visual consistency. It documents the key elements of the system, from geometric construction and proportions to visual appearance and export.

Introduction

An Open and Extensible Project

Although Pure Icons was initially created for KLASSE OS, its use is not limited to this platform. The project is designed to be applicable across other digital environments as well.
Its open-source approach allows designers and developers to use, adapt, and further develop the icon set, subject to the terms of the project's license. The shared design principles aim to preserve the system's visual unity as it evolves.

Introduction

How to Use This Guide

Begin by familiarizing yourself with the design philosophy and principles to understand the thinking behind the Pure Icons visual language. The following sections introduce the rules for creating icons, including geometric and visual requirements, followed by practical guidance on the design, export, and implementation process.

Design Philosophy & Principles

Visual Identity

The visual identity of Pure Icons is built around a balance between clean forms, a modern appearance, and refined visual depth. Our goal is to create an icon system that is minimalist, distinctive, and easily recognizable.
The design direction draws inspiration from two key sources: Nokia MeeGo iconography and Apple's Liquid Glass design language. MeeGo's clear, simple forms and Liquid Glass's layering, lighting, and spatial effects represent different approaches to visual design. Pure Icons treats these inspirations as starting points for an independent visual concept.

Design Philosophy & Principles

Visual Identity

The system is built on simple, clearly defined forms. These are enhanced with lighting effects, shadows, reflections, and layered elements to create visually rich icons without sacrificing clarity.

The objective is not to choose between flat and dimensional representation, but to combine the two deliberately. Simple forms ensure recognizability, while depth and lighting give the icons their distinctive character.

Pure Icons does not aim to copy existing icon styles. Instead, it reinterprets its inspirations through independent design decisions, bringing them together in a consistent and extensible visual system.

Design Philosophy & Principles

Simplicity & Recognizability

The primary purpose of an icon is to communicate information visually. A well-designed icon should be understood quickly while retaining the distinctive character of Pure Icons.
The design process should prioritize essential visual elements. Unnecessary details, overly complex silhouettes, and decorative elements that interfere with understanding should be avoided.
Minimalism, however, does not mean a completely flat appearance or the removal of all detail. In Pure Icons, visual richness comes from the deliberate use of form, layering, and material effects. Every detail should contribute to the icon's character, dimensional appearance, or clarity.
Icons should be evaluated both individually and as part of the complete set. Checking recognizability and visual balance at different sizes helps prevent detailed designs from losing their impact when displayed at smaller scales.

Design Philosophy & Principles

Layering & Visual Depth

Layering is one of the defining visual characteristics of Pure Icons. It gives icons a more sophisticated appearance than traditional flat representations while preserving the simplicity of their underlying forms.
Several visual techniques contribute to the perception of depth:

- **Layers:** Overlapping or nested surfaces that distinguish different elements of an icon.
- **Lighting & highlights:** Light effects that emphasize contours and establish the dimensional character of surfaces.
- **Shadows:** Visual cues that communicate the distance between layers and separate the icon from its background.
- **Material effects:** Techniques such as transparency and reflections that help define the character of individual surfaces.

Design Philosophy & Principles

Layering & Visual Depth

These techniques must be used as parts of a coherent system. They should never be purely decorative: layering should support composition, lighting should clarify form, and shadows should communicate spatial separation.
The objective is to balance visual depth with simplicity. Excessive or inconsistent effects can undermine the unity of the icon set, so their application must follow the rules established in the technical sections of this guide.

Design Philosophy & Principles

Visual Consistency

Pure Icons is a unified icon system, not merely a collection of unrelated illustrations. Individual icons may represent different concepts and use different color palettes, but they must share common geometric and visual principles.

Consistency is expressed through proportions, composition, layer structure, lighting and shadow effects, and the treatment of details.

Consistency does not require every icon to have the same structure or visual weight. Different symbols may require their own forms and character. The design system ensures that these differences remain part of a shared visual language.

When creating new icons, designers should therefore evaluate not only the quality of the individual icon but also how well it fits into the existing set.

Design Philosophy & Principles

Balancing Function & Aesthetics

The visual sophistication of Pure Icons must never come at the expense of usability. Icons should be both aesthetically refined and functionally effective.
Color, form, depth, and material effects should support visual communication. They must not obscure the symbol, conceal essential details, or introduce unnecessary visual noise.
The entire composition must be considered during the design process. An individually striking detail does not necessarily improve the icon as a whole. The best solution is one in which form, function, and visual appearance reinforce one another.
Usability must also be evaluated at different display sizes. Additional detail is justified only as long as it strengthens the icon's character without compromising its clarity.

Design Philosophy & Principles

Openness & Extensibility

Pure Icons was originally created for KLASSE OS, but the project is not limited to a single platform. The icon set is intended to be usable and adaptable across different digital environments, including Linux, Windows, and Android.
The project's open approach allows anyone to explore, use, and further develop the icon set according to their own ideas, subject to the terms of the chosen license. The set can serve as a foundation for custom interfaces, operating systems, or entirely new visual concepts.
We do not intend to prescribe the direction in which Pure Icons must evolve. Anyone who wishes to adapt its visual language to a new platform, introduce new ideas, or take the project in a different direction should be able to do so.

Design Philosophy & Principles

Openness & Extensibility

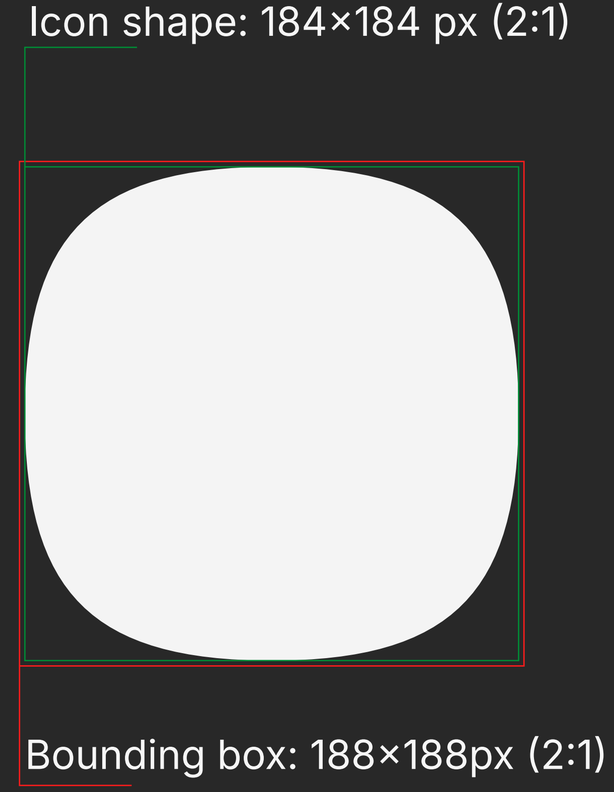
The shared principles and documentation help contributors understand the original system while leaving room for further development. Users can adapt the icon set to their own needs, allowing both the original Pure Icons system and its derivatives to continue evolving.
The goal of Pure Icons is not simply to release a finished icon set, but to provide a foundation from which others can create their own solutions.
One technical detail remains to be finalized in Chapter 08: the project's license. It will determine the conditions under which users can modify, redistribute, and incorporate the icon set into their own projects.

Geometric System

Base Size & Coordinate System

The original design size of Pure Icons is 184 × 184 px. This square canvas provides a common starting point for defining the geometric structure and internal composition of each icon. The base size establishes a consistent reference for constructing the outer shape, positioning internal elements, and defining the spatial relationships between them. Icons are created in vector format, allowing them to be scaled to different sizes while preserving the underlying source geometry.

The 184 × 184 px canvas defines the original design dimensions; it does not mean that icons must be displayed exclusively at this size across all platforms.



Base Shape: Squircle

The base shape of Pure Icons is a mathematically defined, symmetrical rounded square constructed from four cubic Bézier curve segments. This shape establishes the consistent outer geometry of the icon set.

Geometric Specifications

- **Nominal dimensions:** 184 × 184 px
- **Construction symmetry:** A symmetrical curve system consisting of four axis-aligned anchor points and their associated control points
- **Curve type:** Cubic Bézier curve
- **Number of curve segments:** 4
- **Center point:** (92, 92)
- **Bounding dimensions:** 184 × 184 px

Anchor Points

The four primary anchor points are positioned at the midpoints of the shape's sides:

Anchor points	X	Y
Left	0	92
Right	184	92
Top	92	0
Bottom	92	184

The curve segments connecting these anchor points define the rounded contour. The positions of the control points determine the shape of the transitions, while their symmetrical arrangement maintains the geometric balance of the form.

Base Shape: Squircle

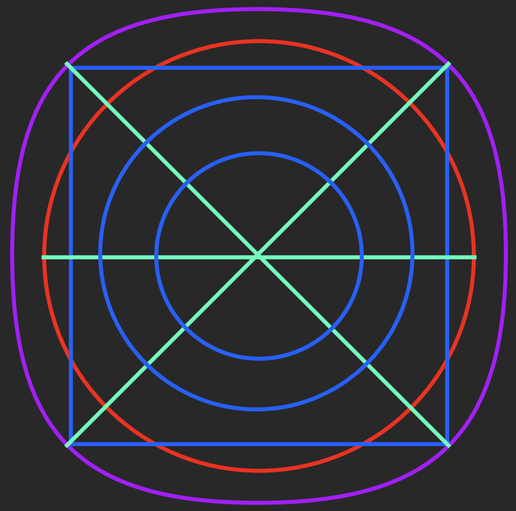
Mathematical Construction

The general equation of a cubic Bézier curve is: $B(t)=(1-t)^3P_0+3(1-t)^2tP_1+3(1-t)t^2P_2+t^3P_3$
where t E [0,1], P_0 and P_3 are the endpoints of the curve segment, and P_1 and P_2 are its control points.
The precise shape of the Pure Icons curves is determined by the points and control points defined in the source vector. Preserving these coordinates ensures that the shape can be reproduced consistently throughout the icon set.

Geometric Reproduction

When modifying the base shape, its square bounding dimensions, symmetry, and the relationships between the four curve segments must be preserved. When scaling icons, the vector geometry should be scaled proportionally.
The Pure Icons squircle is defined by its Bézier construction in the source vector. Therefore, the actual path and its control points constitute the authoritative geometric specification for reproduction.

Grid & Internal Layout



Grid Structure

The Pure Icons grid system establishes a shared geometric framework for positioning and scaling the internal content of icons. It defines maximum content boundaries, recommended alignment references, and a minimum content size while preserving flexibility in individual icon design. The grid consists of five concentric or aligned geometric references, each serving a distinct purpose.

Color	Geometry	Dimensions	Purpose
Purple	Outer squircle	184×184 px	Defines the outer shape of the icon.
Red	Inner circle	164×164 px	Defines the maximum content boundary. Icon content **must not extend** beyond this circle.
Blue	Square	144×144 px	Provides an alignment reference for square-shaped elements.
Blue	Outer circle	120×120 px	Recommended reference size for shapes placed within the icon.
Blue	Inner circle	80×80 px	Defines the minimum content size. Individual icon content **must not be smaller** than this reference.

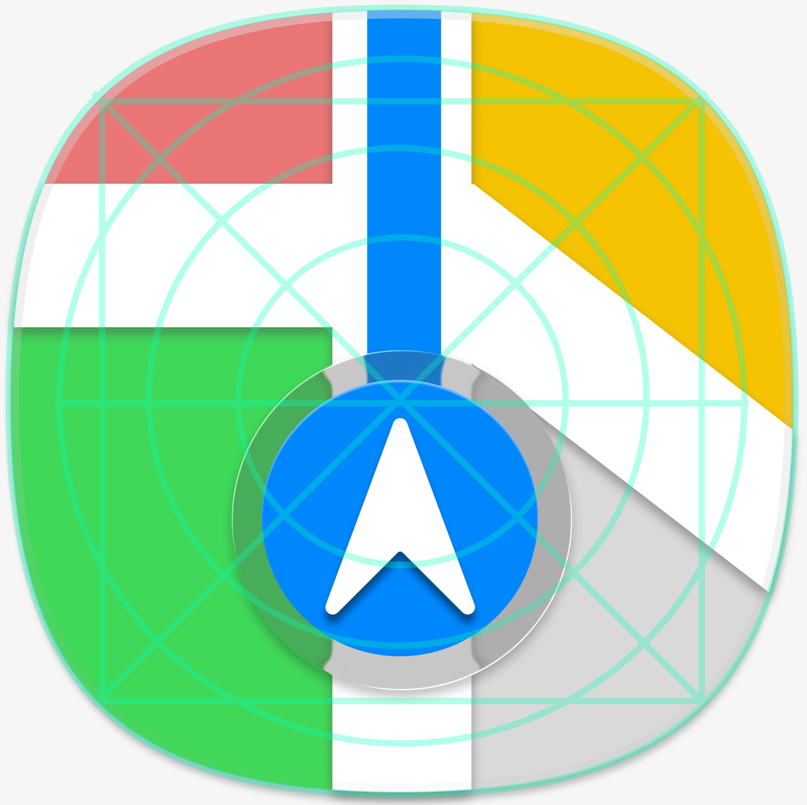
Grid & Internal Layout, Recommended Grid Usage

The 144 × 144 px square and the 120 × 120 px circle are the primary alignment references for icon construction. They provide a consistent starting point without imposing a single mandatory size on every internal symbol.
Designers may adapt the size and proportions of individual elements to suit the symbol being represented. However, the content must remain within the 164 × 164 px boundary, and the 80 × 80 px reference establishes the minimum permitted content size.
This approach balances consistency with creative freedom. Rather than prescribing the exact dimensions of every internal shape, the grid defines clear limits and recommended references that help maintain visual coherence across the icon set.



Grid & Internal Layout, Recommended Grid Usage





Base Shape: Squircle

The Pure Icons base shape is a predefined vector element that establishes the outer geometry of each icon. It uses the standard 184 × 184 px dimensions and provides the foundation for all subsequent layers.
The base shape is supplied as a reusable asset. Designers must use the provided vector source rather than reconstructing the geometry for each new icon. This ensures consistent proportions throughout the icon set.



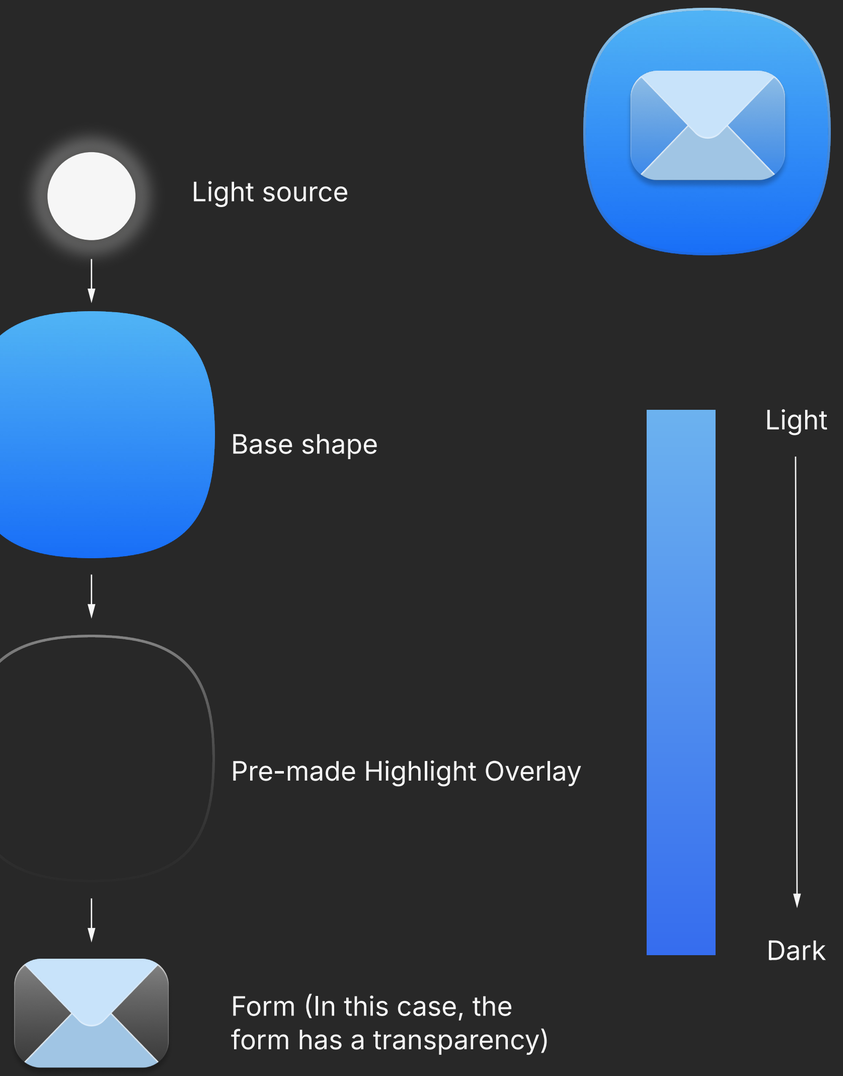
Base shape 184×184 px (2:1)
Base color: f3f3f3



Highlight184×184 px (2:1)
Premade item

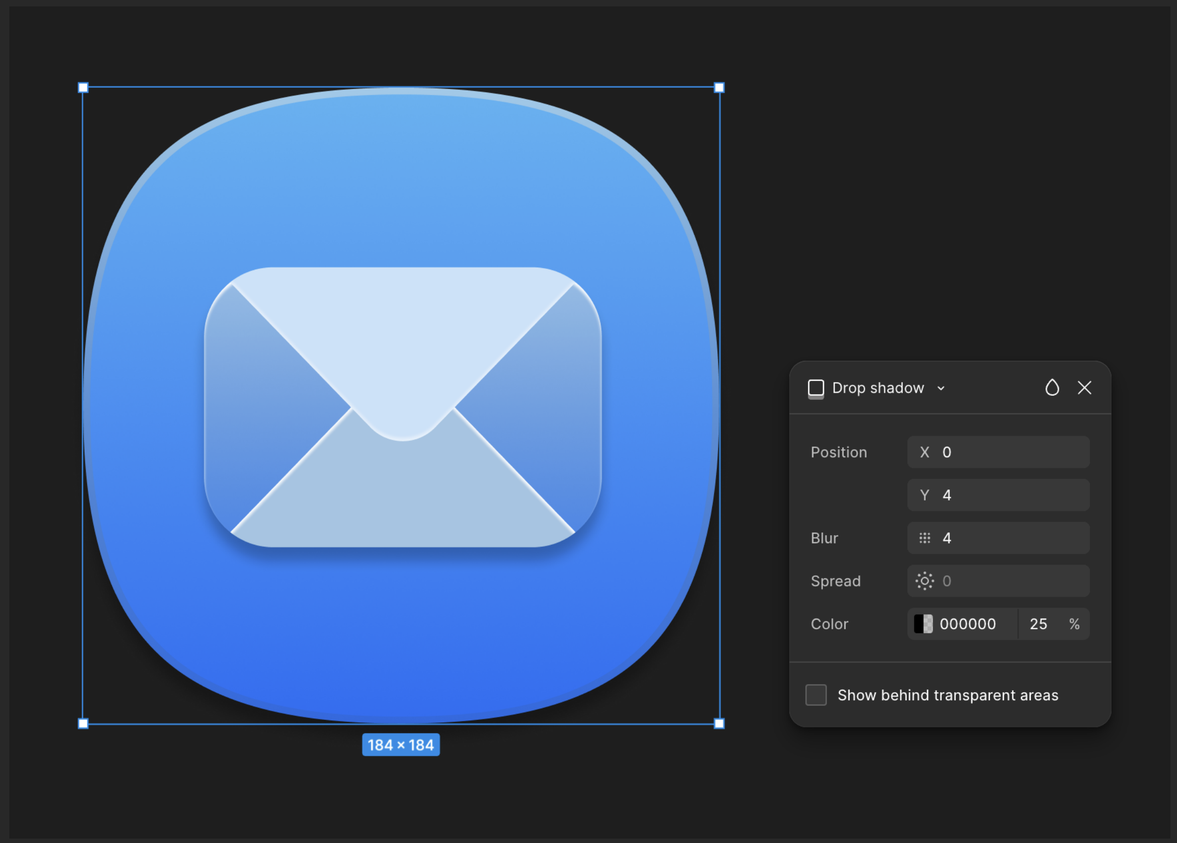
Gradient & Lighting

The base shape uses a smooth color gradient to establish its visual appearance. The gradient transitions from a lighter tone at the top to a darker tone toward the bottom, creating a natural sense of illumination and depth.
The gradient is designed around a consistent light direction: the upper region receives more light, while the lower region appears darker. This relationship gives the base shape a dimensional appearance while preserving its clean, rounded silhouette.
The gradient should remain smooth and visually balanced. Abrupt transitions or inconsistent lighting directions should be avoided to maintain a coherent appearance across the icon set.



Global Lighting & Shadow

The base shape uses a predefined drop shadow to create visual separation from the background and establish a subtle sense of elevation.
The shadow is configured with a vertical offset of 4 px, a blur radius of 4 px, and a spread of 0 px. Its color is black (#000000) at 25% opacity, with no horizontal offset.
These values are defined in the Pure Icons Figma specification and must be preserved for consistency across the icon set.
Implementation rule: Use the specified shadow values without introducing arbitrary variations. Any future changes should be made to the shared specification rather than independently for individual icons.
One distinction worth preserving in the guide: these values belong specifically to the base shape. If the inner plate or individual glyphs use separate shadows, document those as distinct effects with their own Figma values.



Global Lighting & Shadow

The inner form uses a predefined drop shadow to create depth and visual separation from the underlying base shape.
The effect uses a vertical offset of 4 px, a blur radius of 4 px, and a spread of 0 px. The shadow color is black (#000000) at 25% opacity, with no horizontal offset.
These values are defined in the Pure Icons Figma specification and must be preserved when applying the effect to the form.
Note: The selected form also has an Inner Shadow effect enabled in Figma. Its parameters are not visible in this screenshot, so they should be documented separately once its settings are available.



Global Lighting & Shadow

The inner form uses a subtle white inner shadow to create a surface highlight and enhance its dimensional appearance. The effect is configured with an X offset of 0 px, a Y offset of 1 px, a blur radius of 1 px, and a spread of 0 px. The color is white (#FFFFFF) at 100% opacity.
While these values define the standard effect, designers retain some creative freedom in how the highlight contributes to the appearance of individual forms. The result should remain visually coherent with the Pure Icons lighting direction and overall style.
Design principle: The effect settings provide a consistent starting point, while the final visual balance can be adjusted to suit the shape and material being represented.



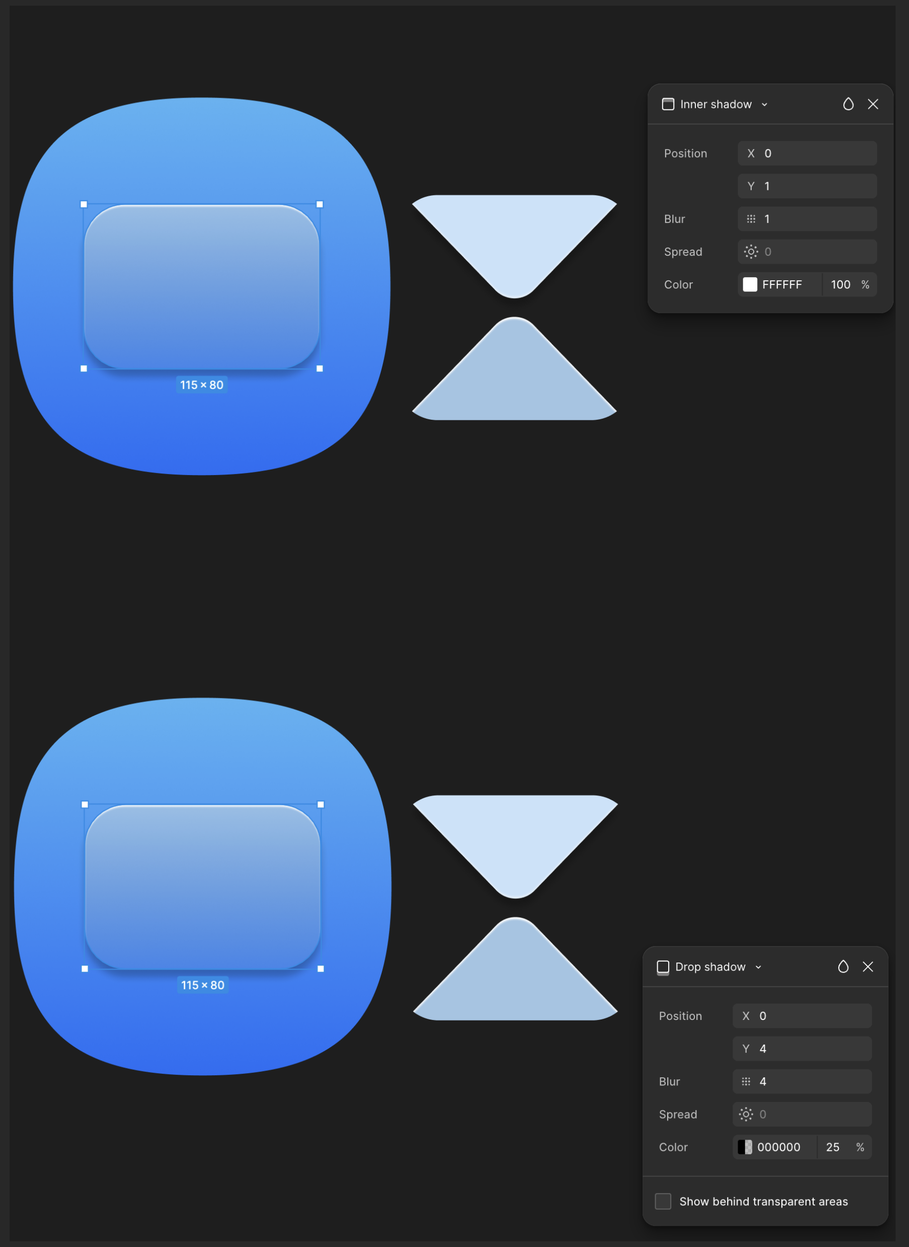
Inner Plate

The inner form consists of three overlapping shapes that work together to create the layered, glass-like appearance of the icon. Each layer follows the lighting principles defined in the previous section, with effects applied according to its position and function.

Base Form

The base form is a rectangle measuring 115 × 80 px, compliant with the Pure Icons grid system. It serves as the foundation for the inner construction.

The base form uses the established inner white highlight and drop shadow settings. A vertical transparency gradient, transitioning from top to bottom, creates the impression of a



Inner Plate

Upper Layers

Two additional shapes are placed over the base form. Both follow the same lighting principles, but their effects differ according to their position in the construction.
The middle layer uses an inner highlight and a drop shadow to establish separation from the underlying surface.
The top layer uses the inner highlight to define its illuminated edge. It does not use a separate drop shadow, as the base form already provides the overall shadow for the construction.

Layering Principle

Lighting effects should be applied selectively rather than duplicated across every layer. Each effect must support the perceived depth and spatial relationship between the shapes.
This approach maintains a coherent lighting direction, prevents unnecessary shadow accumulation, and preserves the refined glass-like appearance of Pure Icons.

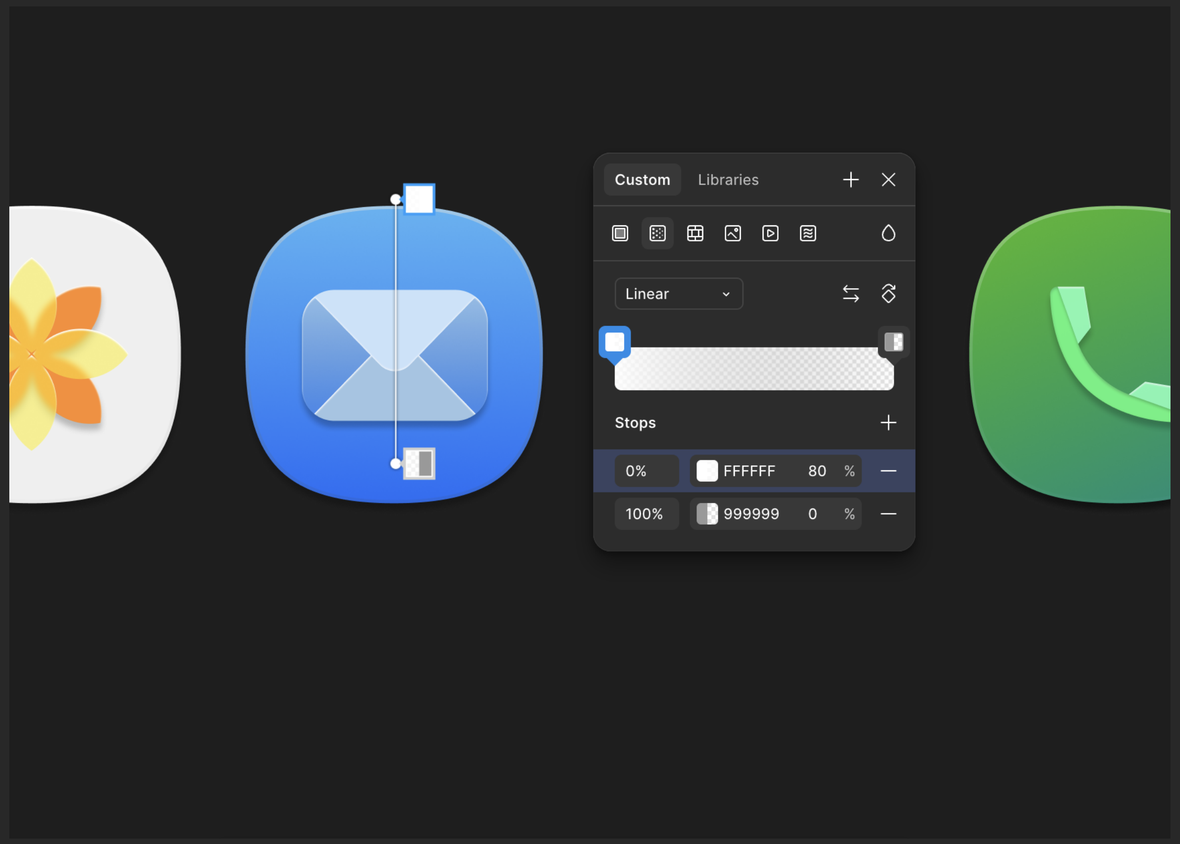


Content & Materials

The material appearance of Pure Icons is created through a combination of transparency, blur, highlights, and shadows. Different combinations of these effects are used to achieve glass-like and plastic-like surfaces.

Transparency & Glass Effect

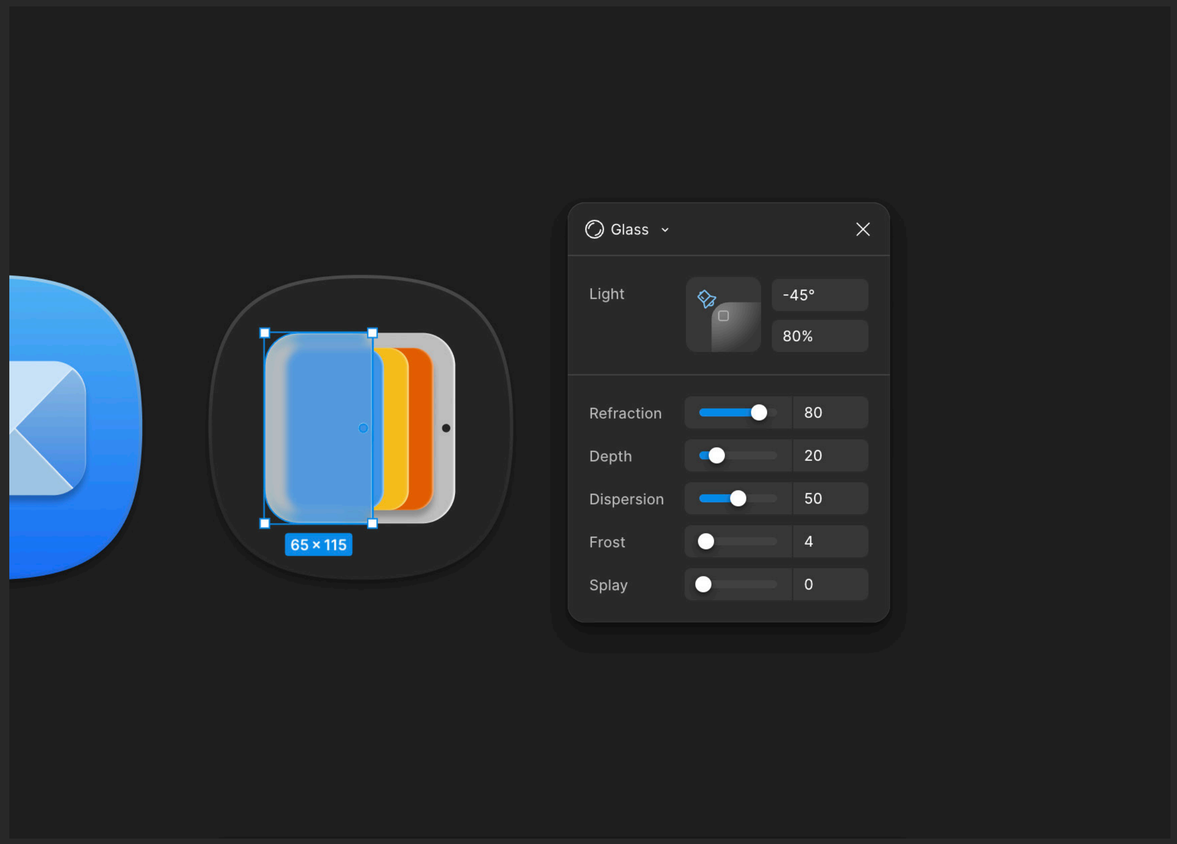
In the Mail icon, transparency is used to create a glass-like appearance. A gradual transparency transition allows the underlying surfaces to influence the appearance of the shape, adding depth while preserving visual clarity.



Content & Materials

(Liquid) Glass

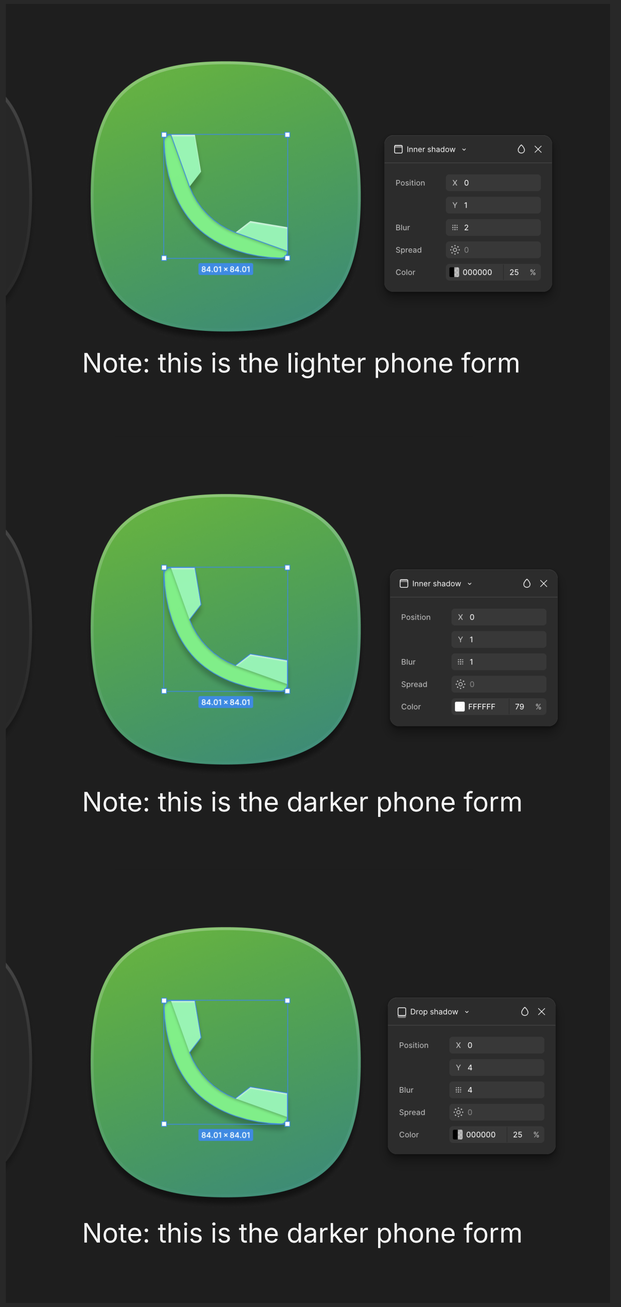
The Wallet icon uses Figma's built-in Liquid Glass template to achieve its distinctive material appearance. This approach introduces a different surface treatment while remaining compatible with the overall visual language of Pure Icons.



Content & Materials

Three-Dimensional Surface Effects

The Phone icon uses highlights and drop shadows to create a three-dimensional glass-like or plastic-like appearance. These effects define the edges and surfaces of the symbol, giving it a raised, dimensional quality.



Note: this is the lighter phone form

Note: this is the darker phone form

Note: this is the darker phone form

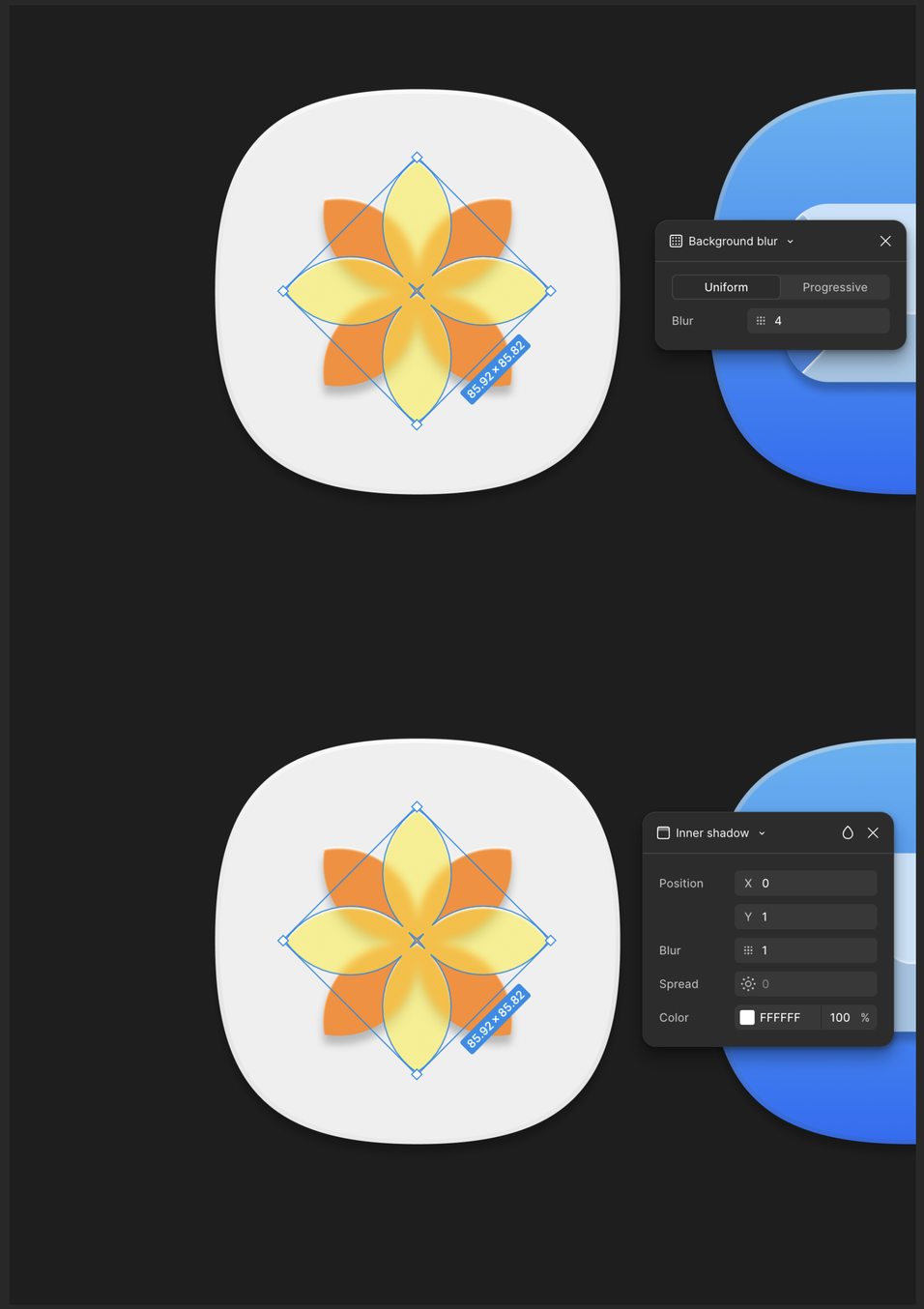
Content & Materials

Transparency & Layer Blur

The Gallery icon combines transparency with blur applied to individual layers to achieve its glass-like appearance. The interaction between translucent surfaces and blurred layers creates depth and a softer visual transition between overlapping elements. The transparency is set to 50%.

Material Consistency

These combinations of surface treatments form the material language of Pure Icons. Individual icons may use different techniques depending on the intended appearance, while maintaining a consistent approach to lighting, depth, and visual refinement.



Layering Rules

Pure Icons are constructed from multiple graphical layers. The correct layer order, consistent application of lighting effects, and clear visual separation between elements establish the icons' dimensional appearance and maintain a unified visual language.

Layer Order

Icon construction follows the four phases defined in the previous sections:

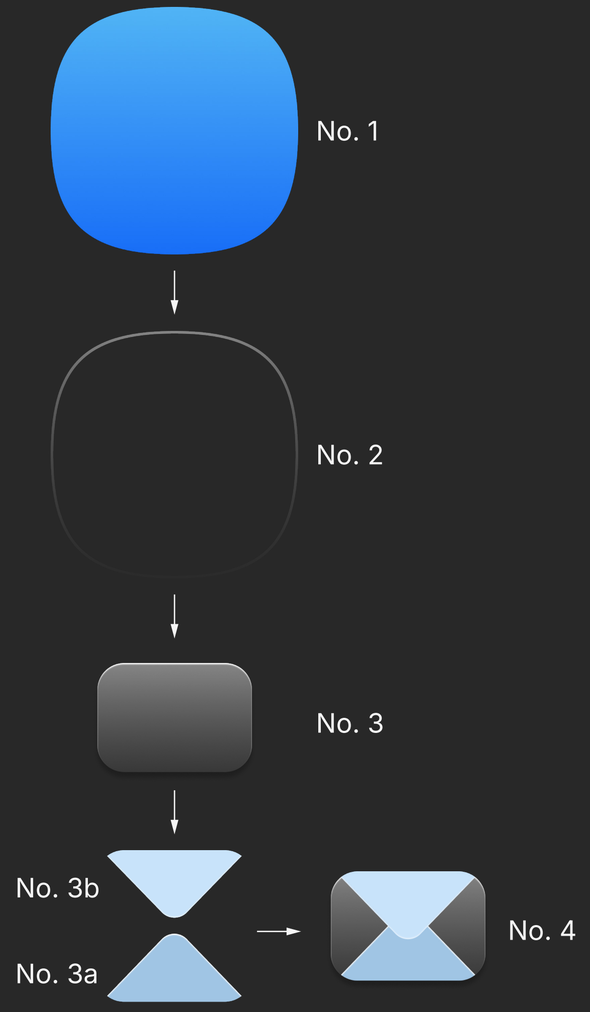
1. Base Squircle: establishes the background and primary shape of the icon.

2. Global Lighting & Shadow: defines the base shape's lighting and separates it visually from the surrounding background.

3. Inner Plate: consists of the internal graphical layers that form the icon's inner surface.

4. Content & Materials: defines the visual character of the icon through surface properties such as transparency, blur, highlights, and shadows.

Each phase may contain multiple graphical elements. The inner form, for example, can consist of several overlapping shapes, each serving a distinct visual purpose.

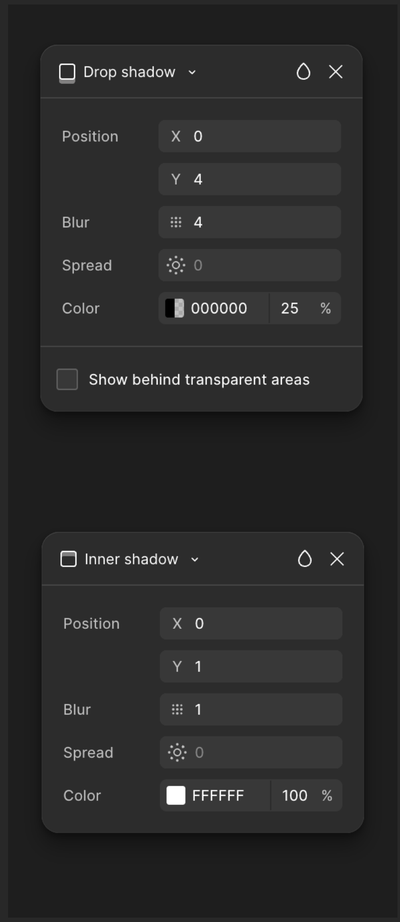


Layering Rules

Consistent Application of Lighting Effects

Lighting and shadow effects must follow a consistent lighting logic. The effects applied to each layer should reflect its spatial position and visual function.

- The base shape uses the predefined drop shadow.
- Inner forms may use their own highlights and, where necessary, drop shadows.
- The same effects do not need to be applied to every layer.
- Shadows must be managed carefully to prevent unnecessary accumulation between overlapping layers.
- Previously defined effect settings provide the starting point. Adjustments are permitted when necessary to achieve the appropriate visual balance for the shape and material being represented.



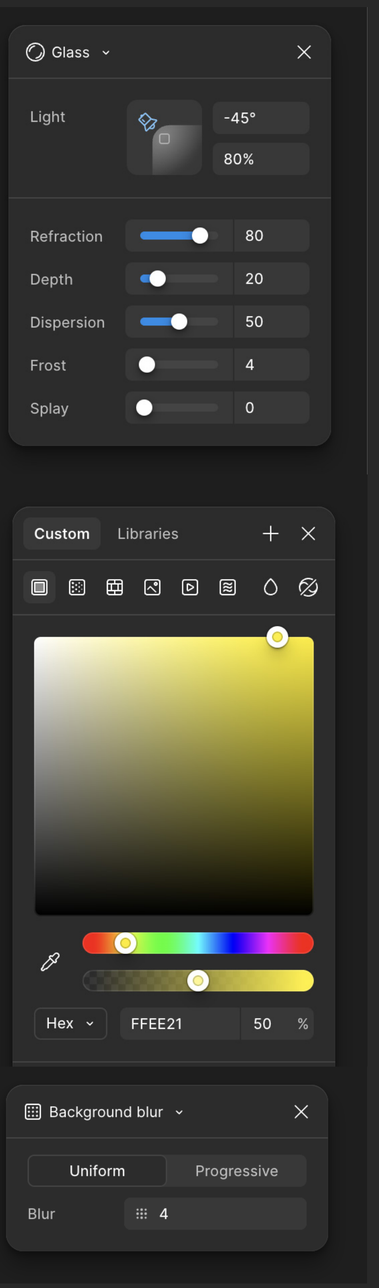
Layering Rules

Spatial Separation

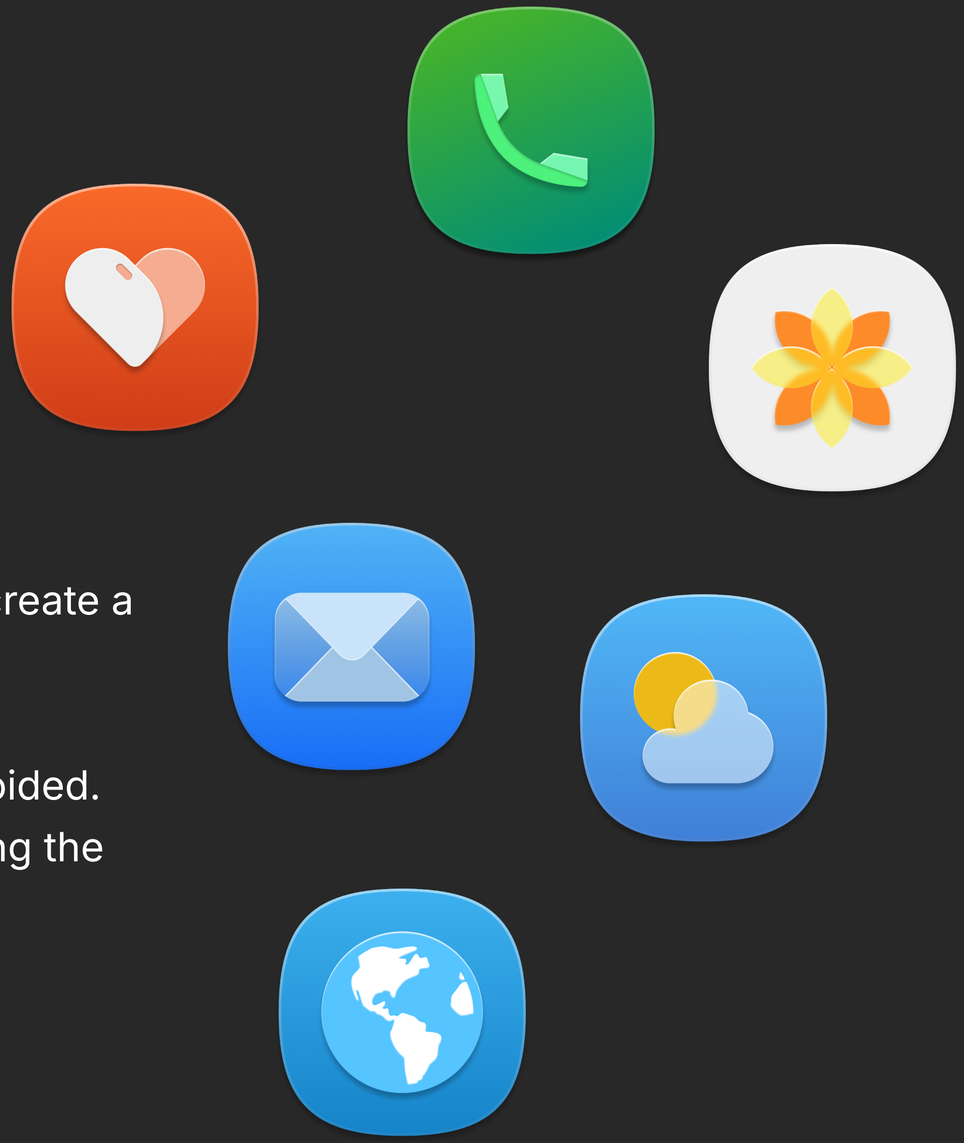
The spatial relationships between layers are communicated through the combined use of highlights, drop shadows, and material effects.
Shadows and highlights should emphasize the relative positions of the shapes. These effects are not decorative elements in isolation; their purpose is to reinforce the perception of depth and spatial structure.

Material Effects & Layer Relationships

Transparency, blur, and surface highlights contribute to the material appearance of an icon. These effects should be combined according to the intended visual result. Different icons may use different material treatments, but the underlying principles of lighting, depth, and visual refinement must remain consistent throughout the Pure Icons system.



Layering Rules



Visual Balance

The objective of layering is not to apply as many effects as possible, but to create a clear, balanced, and spatially coherent icon.

Every layer and effect should serve a distinct visual purpose. Unnecessary shadows, excessive highlights, and competing surface effects should be avoided.

The result should preserve the recognizability of the symbol while maintaining the refined, dimensional appearance characteristic of Pure Icons.

Colour & Gradients

Colour Selection

The colour of each Pure Icons icon is determined by its function and visual identity. The project does not enforce a universal colour palette, allowing individual icons to use colours that best represent their purpose.

Gradients & Lighting

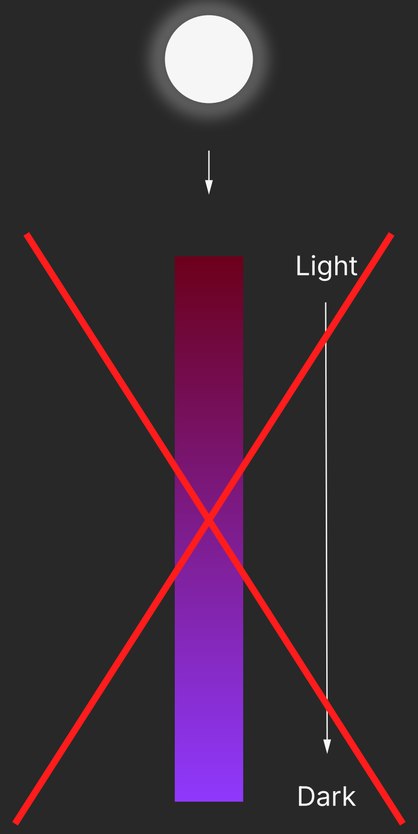
Gradients are a fundamental part of the Pure Icons visual language. Each icon must use a gradient to support the lighting effect and create a sense of depth across its surfaces.
The gradient should establish a smooth transition between lighter and darker areas, reinforcing the perception of illumination and contributing to the dimensional appearance of the icon.



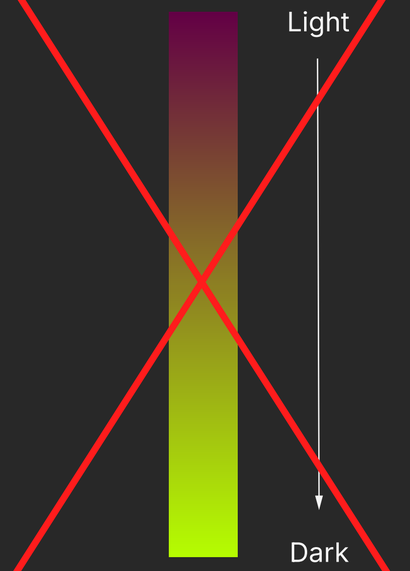
Colour & Gradients

Visual Consistency

Although colours may vary between icons, gradients and lighting effects must remain visually coherent with the overall design language of Pure Icons. Colour selection is flexible, but the use of gradients to support lighting is a defining characteristic of the system.



Inconsistent coloring, the direction of the gradient cannot be adjusted.



It is also important to consider color harmony.

Icon Creation Workflow

Step by step

1. Preparation
Icon creation begins with defining the function to be represented and establishing its visual concept. The symbol must communicate the intended function clearly while remaining consistent with the visual language of Pure Icons.

2. Geometric Construction
The symbol is constructed according to the geometric system defined in Section 03 — Geometric System. The design must follow the established base size, grid system, proportions, and negative space principles.
The objective is to create a recognizable, balanced shape that remains legible at different display sizes.

3. Layer Assembly
Once the geometry is established, the symbol is integrated into the Pure Icons layer architecture. This involves using the predefined base shape and top-edge highlight, then assembling the layers that form the inner structure.
The order and relationships of these layers must follow the rules defined in Section 04 — Layer Architecture.

Step by step

4. Lighting & Material Effects

The next step is to establish the lighting and material appearance. Previously defined effect settings serve as the starting point, while transparency, blur, highlights, and drop shadows are applied according to the visual requirements of the design.

The objective is to achieve the intended depth and material appearance while preserving visual consistency across the icon system.

5. Visual Review

The completed icon must be evaluated against the principles of Pure Icons. The review should consider:

- The recognizability and legibility of the symbol.
- Geometric proportions and visual balance.
- The consistency of lighting and shadow effects.
- The harmony of colours and material treatments.
- The icon's compatibility with the overall Pure Icons visual system.

If the icon does not meet these requirements, the relevant design stages should be revisited and adjusted.

Step by step

6. Export Preparation

Once the icon has been finalized, it must be prepared for export and implementation. Export formats, file naming conventions, and technical requirements are documented in Section 07 — Export & Implementation.

Export & Implementation

Export Format

The primary export format for Pure Icons is SVG (Scalable Vector Graphics). This vector format allows icons to be scaled without loss of quality while preserving their geometric shapes and supported visual effects.
Icons must be exported as SVG files to ensure they can be used across digital interfaces and compatible applications.

Transparent Background

Exported icons must not contain unnecessary solid background fills. The base shape is part of the icon, but the area outside it must remain transparent.

File Naming & Organization

Icon filenames must be short, descriptive, and consistent. Filenames should use lowercase letters, with words separated by hyphens where necessary.
Examples:
- mail.svg
- phone.svg
- gallery.svg
- wallet.svg

Exported files must be stored in an organized directory structure, separate from editable source files.

Export & Implementation

SVG Compatibility

SVG files must preserve the shapes, gradients, transparency, blur effects, and shadows required for the intended appearance of each icon.

After export, every icon must be checked in an SVG-compatible viewer. Particular attention should be paid to translucent surfaces, layer blur effects, and drop shadows, as their rendering may vary between environments.

Quality Assurance

Before publication, verify that:

- The icon renders correctly as an SVG.
- Gradients and lighting effects display as intended.
- The transparent background is preserved.
- The icon remains recognizable at different display sizes.
- The file opens correctly and contains no export errors.

Any inconsistencies should be resolved by adjusting the export settings or correcting the SVG file.

Export & Implementation

7.6 Implementation

Pure Icons SVG files can be used in compatible digital interfaces and applications. During implementation, verify that the target environment supports the SVG effects used in the icon.

If a platform does not render a particular effect correctly, its compatibility must be investigated in that environment.

Preserving the original vector source ensures that future modifications remain possible.

Resources & Contributions

Project Resources

The Pure Icons project resources include the icon set, editable source files, and design system documentation. These resources are intended to support the use, understanding, and further development of the project.
The GitHub repository serves as the central location for the project, providing access to the icons, related files, and documentation. The files must be organized so that users can easily find the resources they need.

Contribution Guidelines
Pure Icons is an open project that allows the community to improve existing icons and create new ones. Contributors must follow the geometric, layering, and visual principles defined in this Design Guide. Every new icon must align with the consistent visual language of Pure Icons.
Submitted icons must be recognizable, visually balanced, and technically usable. Where necessary, submissions should include editable source files and an exported SVG file.
Contributions are subject to review before being incorporated into the official icon set. This ensures that quality and visual consistency are maintained as the project evolves.

Design Consistency

Preserving the shared design principles is essential as Pure Icons continues to evolve. New icons and modifications to existing ones must not compromise the system's recognizable visual identity.
Geometric proportions, lighting, gradients, layers, and material effects must remain consistent with the documented principles.
The system is extensible, and individual icons may use different visual treatments according to their intended function. However, these variations must not compromise the unified visual language of Pure Icons.

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